Practice workbook · Local and global variables

Defining local and global variables, where each is created and where each can be accessed, and contrasting them.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T2-03

1. Question 1 · part e · 2 marks

2019 JUN · Paper 02

Worked-example retry · Attempt part e only.

1 Computer programs use several different programming constructs.

(a) Identify the **most appropriate** data type for the value 4.5

(1)

- ☒ **A** Boolean
- ☒ **B** Character
- ☒ **C** Real
- ☒ **D** String

(b) Give a definition of a logic error.

(1)

(c) Open **Q01c** in the code editor.

The program should print out a counter and the counter with 7 added to it.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

(d) Give **one** reason why computer programmers should use comments in their programs.

(1)

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1. Question 1 · part e · 2 marks

2019 JUN · Paper 02

Worked-example retry · Attempt part e only.

(e) Describe the difference between a local variable and a global variable in a computer program.

(2)

(f) Trevor is the manager of a shoe shop.

Sales assistants can earn a bonus based on the numbers of pairs of shoes they sell and the total income for the shop each day.

Open **Q01f** in the code editor.

Amend the code to complete the 'if statement' used to produce the outputs described in the table.

Condition	Output
Shop income is more than £5000 or sales assistant has sold at least 10 pairs of shoes	Bonus is 10% of salary
Shop income is £2000 or more and sales assistant has sold at least 5 pairs of shoes	Bonus is 5% of salary

Do not add any further functionality.

Save your amended code as **Q01fFINISHED** with the correct file extension for the programming language.

(4)

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P 6 1 8 8 1 R A 0 3 1 6

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Turn over

1. Question 1 · part e · 2 marks · Working and answer

Working and answer

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2. Question 2 · part a.iii · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part a.iii only.

2 Solutions to problems are made up of many different components.

(a) Programmers use subprograms when developing code.

(i) Give **two** reasons to use **subprogram libraries** when developing code.

(2)

1 .....

.....

2 .....

.....

(ii) Identify which **one** of these must return a result.

(1)

- ☒ **A** Function
- ☒ **B** Iteration
- ☒ **C** Procedure
- ☒ **D** Selection

(iii) State what is meant by the term **local variable**.

(1)

.....

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2. Question 2 · part a.iii · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part a.iii only.

(b) A school is planning a fish and chip dinner for the students and their families.

(i) Tickets are only sold to families.

A family is one to four adults and one to six children.

A program is being written to help manage ticket sales.

Complete the table to show examples of normal, boundary and erroneous numeric test data for the program.

(3)

	Adult	Children
Normal		
Boundary		
Erroneous		

(ii) The kitchen staff use a program to determine whether they have enough chips or how many more they need to order.

Open **Q02bii** in the code editor.

There are **four** errors in the code.

Amend the code to correct the errors.

Use this test data to help you find the errors.

Chips in stock (kilograms)	Number of adults	Number of children	Expected output
19	100	150	Stocks available
15	100	150	Order 4.0 kilograms

Save your amended code as **Q02biiFINISHED** with the correct file extension for the programming language.

(4)

(Total for Question 2 = 11 marks)

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2. Question 2 · part a.iii · 1 mark · Working and answer

Working and answer

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3. Question 4 · part a.i · 1 mark

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Further practice · Attempt part a.i only.

4 Programmers write code that keeps track of values, is efficient and uses subprograms.

(a) Global variables are different from local variables.

(i) State where a **global** variable is created.

(1)

.....

.....

(ii) State where a **local** variable is accessible.

(1)

.....

.....

(b) An algorithm reports the range that an integer falls within.

Integers from 0 to 39 are in the low range.

Integers from 40 to 69 are in the middle range.

Integers from 70 to 100 are in the high range.

Figure 4 and **Figure 5** show two algorithms that solve this problem.

```
1  SEND 'Enter an integer: ' TO DISPLAY
2  RECEIVE theNumber FROM (INTEGER) KEYBOARD
3  IF (theNumber < 40) THEN
4      SEND 'Low range' TO DISPLAY
5  ELSE
6      IF (theNumber < 70) THEN
7          SEND 'Middle range' TO DISPLAY
8      ELSE
9          SEND 'High range' TO DISPLAY
10     END IF
11 END IF
```

Figure 4

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P 7 5 7 3 7 A 0 8 1 6

3. Question 4 · part a.i · 1 mark · Working and answer

Working and answer

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4. Question 4 · part a.ii · 1 mark

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Further practice · Attempt part a.ii only.

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P 7 5 7 3 7 A 0 8 1 6

4. Question 4 · part a.ii · 1 mark · Working and answer

Working and answer

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